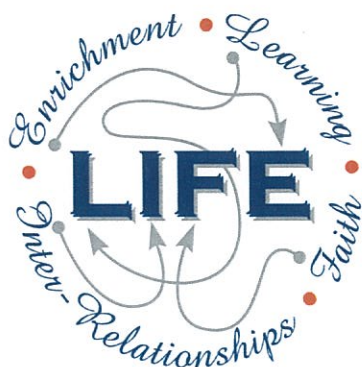




Holy Cross College

Semester 2 Examination 2015

Question/Answer Booklet

**Year 10 Science
(Adjusted)**

Please place your student identification label in this box

Student Name

SOLUTIONS

Student's Teacher

Time allowed for this paper

Reading time before commencing work: 5 minutes
Working time for paper: 50 minutes

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple Choice	13	13	20	13	30
Section Two: Short Answer	9	9	30	31	70
				44	100

Instructions to candidates

- The rules for the conduct of examinations at Holy Cross College are detailed in the College Examination Policy. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer Booklet.
- Working or reasoning should be clearly shown when calculating or estimating answers.
- You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
- Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number.
 - Fill in the number of the question(s) that you are continuing to answer at the top of the page.

SECTION 1 MULTIPLE CHOICE (13 marks)

Choose the best answer for each question in the Multiple Choice Question booklet.

Mark your selection with a cross.

-
- | | | | | |
|----|-----|----------------|----------------|----------------|
| 1 | (a) | (b) | (c) | (d) |
| 2 | (a) | (b) | (c) | (d) |
| 3 | (a) | (b) | (c) | (d) |
| 4 | (a) | (b) | (c) | (d) |
| 5 | (a) | (b) | (c) | (d) |
| 6 | (a) | (b) | (c) | (d) |
| 7 | (a) | (b) | (c) | (d) |
| 8 | (a) | (b) | (c) | (d) |
| 9 | (a) | (b) | (c) | (d) |
| 10 | (a) | (b) | (c) | (d) |
| 11 | (a) | (b) | (c) | (d) |
| 12 | (a) | (b) | (c) | (d) |
| 13 | (a) | (b) | (c) | (d) |

SECTION 2 SHORT ANSWER (25 marks)

Attempt all questions. Write your answers in the spaces provided.

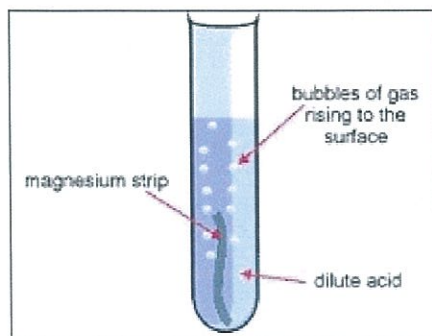
1. In the table below, choose the correct statement for each word or phrase given.
Write the letter of the correct statement next to the word. (7 marks)

WORD	CORRECT LETTER	STATEMENT
Precipitation	D	A - able to be dissolved in water.
Soluble	A	B - chemical reaction where metal is "eaten away" by a substance.
Hazardous substance	G	C - substance formed from decaying plants and animals over millions of years.
Corrosion	B	D - new substance forms when 2 solutions of chemicals are mixed.
Combustion	F	E - substance used to increase the rate of a reaction.
Fossil fuel	C	F - the burning of substances in oxygen.
Catalyst	E	G - something that is harmful to the human body.

2. During the term that you were studying Chemistry, you did an extensive study of one room in your house where chemicals were being stored and/or used.
- (a) Name **one dangerous chemical or substance** you found around your home. (1 mark)
- Name (1)
- (b) Why is this chemical or substance dangerous? What can it do to you? (1 mark)
- Any reasonable answer. (1)
- (c) How do you use the chemical or substance so it will not hurt you or the environment? (1 mark)

• Must mention a safety precaution. (1)

3. It is common to have to speed up reactions in the laboratory to make them produce a new substance as fast as possible. A student set up the following reaction of a strip of magnesium metal in dilute (weak) acid.



Give **two ways** in which you could speed up this reaction.

(2 marks)

- (i) • Add concentrated acid. • Break the magnesium into smaller pieces.
- Add more magnesium
- (ii) • Heat the test tube
- [Any 2 - 1 mark each].

4. Use the listed words to complete the sentences below.

(4 marks)

total	unbalanced	equal	opposite
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An object will remain at rest, or will not change its speed or direction, unless it is acted upon by an outside, unbalanced ($\frac{1}{2}$) force.

The acceleration of an object depends on the mass of the object and the total ($\frac{1}{2}$) force acting on it.

When an object applies a force to a second object, the second object applies an equal ($\frac{1}{2}$) and opposite ($\frac{1}{2}$) force to the first object.

5. Reaction time is the time taken to react to a stimulus or change in surroundings.

List **one thing** that may affect reaction time and explain how it changes the reaction time. (2 marks)

- AGE - improves reaction to 16, then decreases with age.
- ALCOHOL- DRUGS - increases reaction time.
- DISTRACTIONS - increases reaction time. [Any one - 2 marks]
- LEVEL OF CONSCIOUSNESS - fatigue increases reaction time.
- TYPE OF STIMULUS - give a variety of reaction times

6. (a) Explain how a fatigued (tired) driver could cause an accident on the road. (2 marks)

- Lack of attention - misjudge traffic conditions.
- Vehicle can wander on the road.

[Any 2 reasonable causes]

- (b) List **one way** you can reduce fatigue (tiredness) as a driver. (1 mark)

- Plan a good sleep before driving.
- Eat properly.
- Start early in the day.
- Do not have drugs or alcohol.
- Take regular breaks.
- Stay overnight on long trips.
- Share the driving.

7. Explain **one way** that being under the influence of alcohol can affect your ability to drive a car. (1 mark)

- Slower muscle co-ordination and reflexes.
 - Vision and hearing impaired.
 - Brain's ability to receive and understand messages is reduced.
 - Can't judge speed or distances accurately.
 - Slower decision-making.
 - Increased risk taking - speeding.
- [Any one - 1 mark]

8. (a) Name one safety feature of a car. (1 mark)

• Seat belt, air bag, crumple zone, protection cell, padded interior, brakes, tyres. [Any 1]

- (b) Explain (in simple terms) how this safety feature works to keep passengers safe. (2 marks)

• Reasonable explanation (1).

9. In the table below, choose the correct statement for each word or phrase given. Write the letter of the correct statement next to the word. (6 marks)

WORD	CORRECT LETTER	STATEMENT
Reaction time	E	A - how far a car travels in total from the driver seeing a danger and stopping the car.
Braking distance	D	B - when passengers hit the inside of a car as it collides with an object or other car.
Stopping distance	A	C - substance that affects the functioning of the human body.
Drug	C	D - distance covered in a car while applying the brakes to stop it.
Stimulant	F	E - time taken to move or act in response to a stimulus (event).
Second collision	B	F - substance that increases the activity of the brain.

END OF EXAMINATION